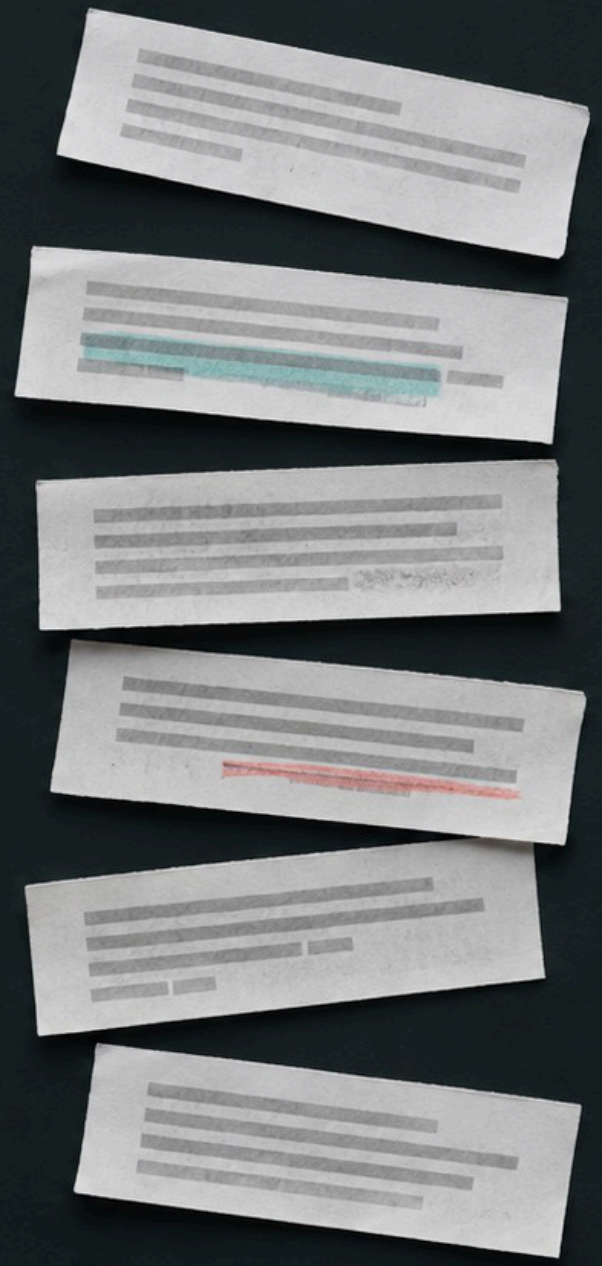
DAMDID 2026

Deep Re-annotation and Controlled Candidate Pools

Re-evaluating ranking models on MS MARCO



A relevant passage can receive zero gain

“what causes boiling point to increase”

Passage (shortened paraphrase)	Original qrels	Human judgment
7112148 Adding salt raises the boiling point.	Relevant	Relevant
7399382 A non-volatile solute raises the boiling point.	Unjudged	**Relevant**
987600 The boiling temperature of acetone.	Unjudged	Not relevant

qrels: query relevance judgments. Unjudged passages receive zero gain.

Data and judgment coverage

English MS MARCO

8.84 million passages

Human judgments

1,179

queries; 33,667 pairs

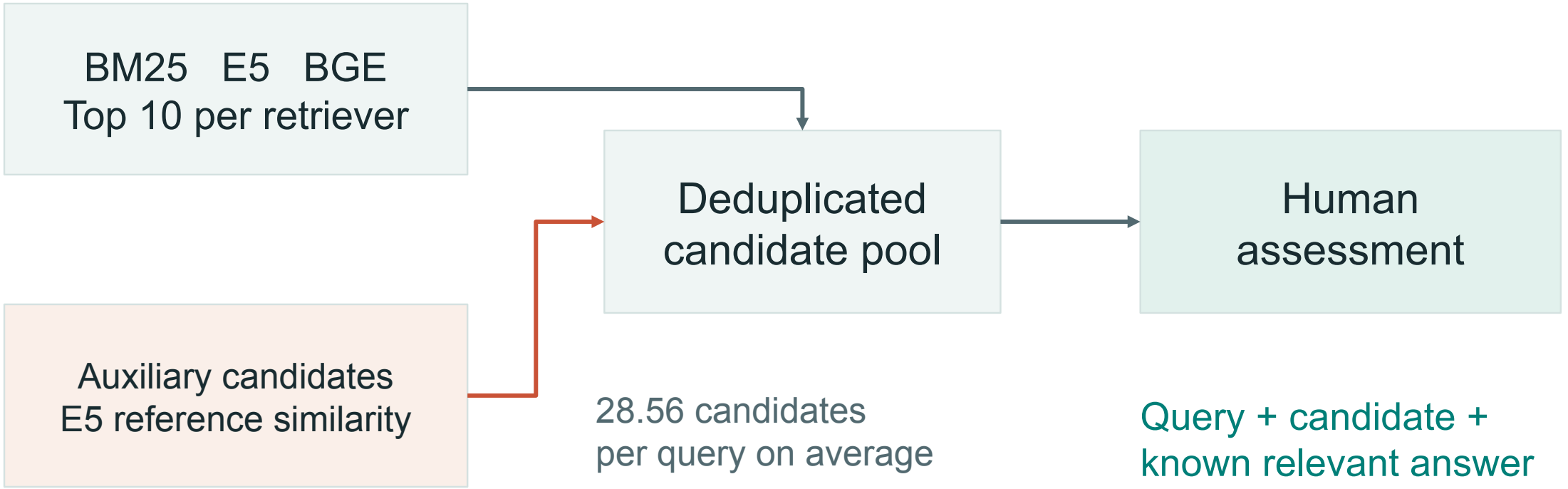
Judge predictions

101,093

queries; varying pool depths

“Deep” describes judgment coverage within selected candidate pools.

Reference-assisted manual annotation



Reference information supports candidate preparation and human assessment.

More relevant judgments within the same pools

Mean passages judged relevant per query

1.05

17.47

Original qrels

1,243 positive pairs

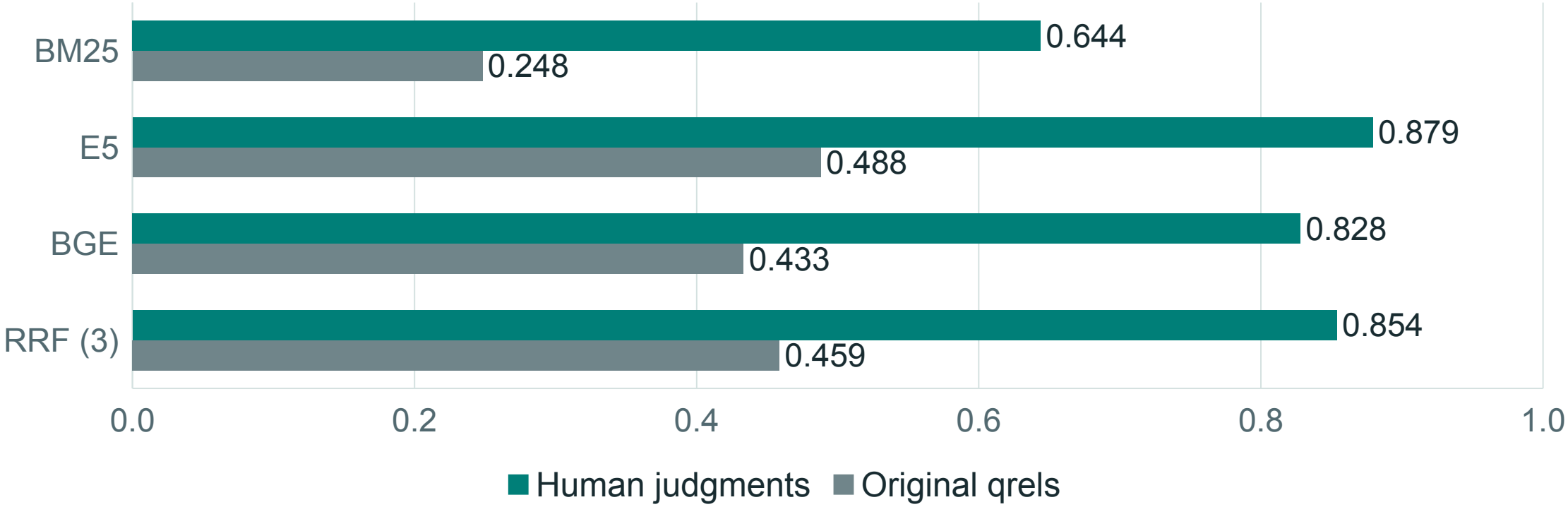
Extended human judgments

20,597 positive pairs

1,179 queries. Same candidate pools and retrieval rankings.

More judgments change measured NDCG@10

RRF: reciprocal rank fusion of BM25, E5 and BGE

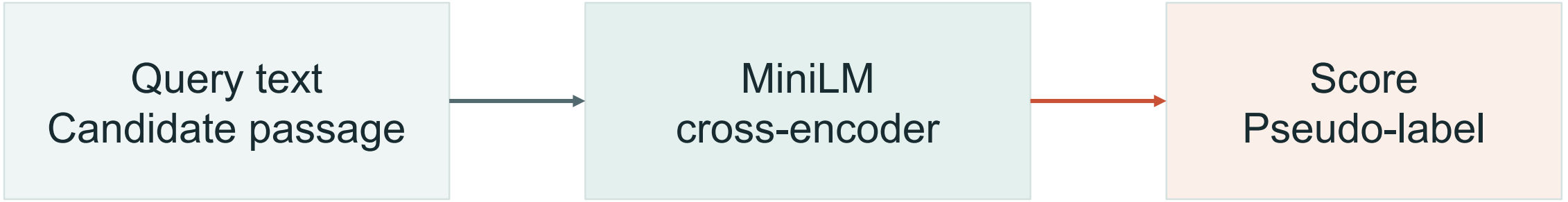


E5 paired change: +0.391 95% CI [0.365, 0.417]

Pooled NDCG@10. Same rankings; shared ideal denominator within each query and label set.

A compact judge extends label coverage

Human labels train a judge for larger candidate pools.



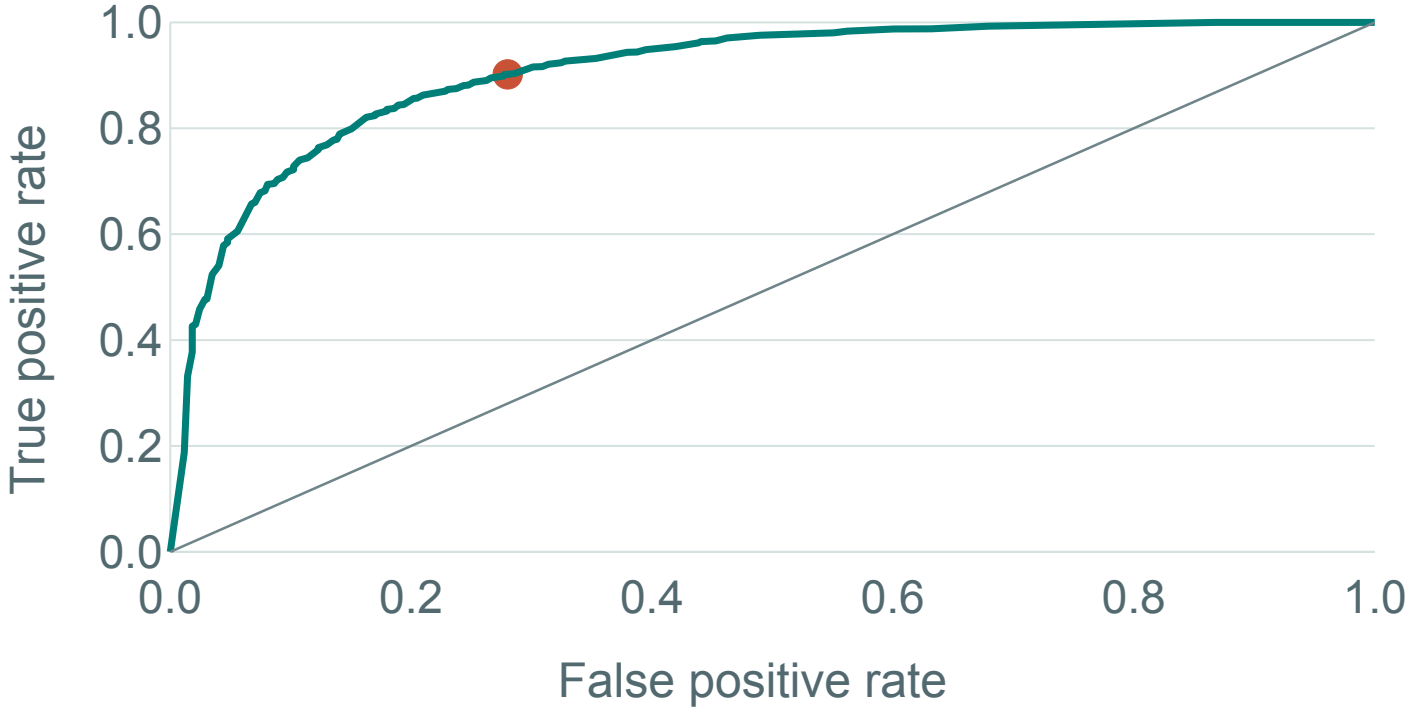
Jointly scores the two texts

Inputs exclude the reference answer, ranks and retrieval scores.

Pseudo-labels are model predictions.

Judge agreement on held-out queries

177 unseen queries from the manual pools, 5,002 pairs



0.907

ROC-AUC

0.875

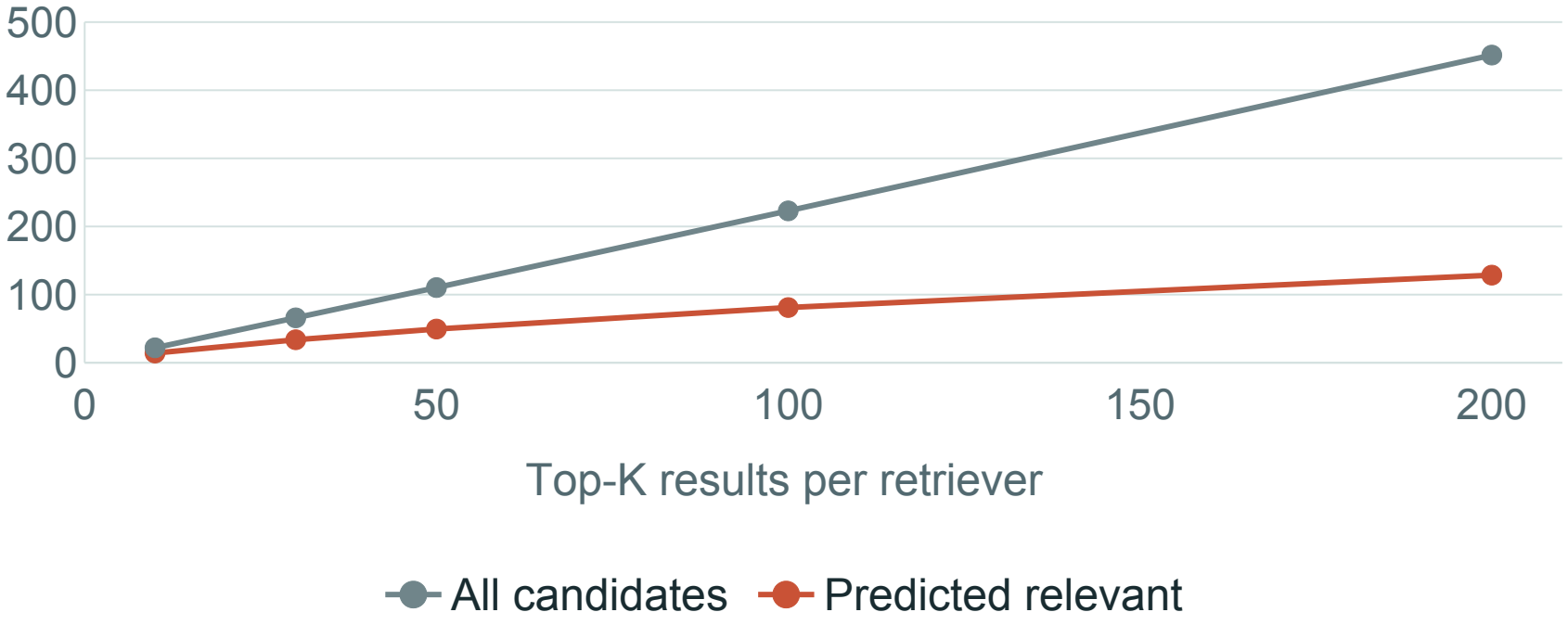
F1 at validation threshold

Marked point: threshold selected on validation queries

Deeper pools have no independent human validation.

Larger pools contain more predicted positives

K: results per retriever used to build the judgment pool



451.5

candidates

128.6

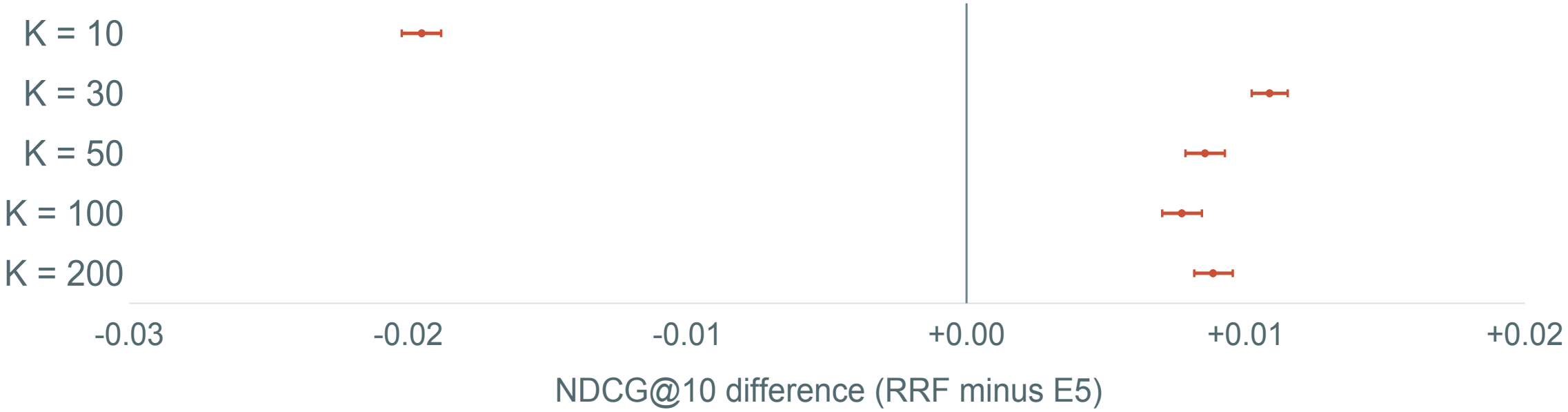
predicted relevant at K = 200

101,093 queries, including zero-positive pools. Same judge and threshold.

Changing the pool and fusion inputs

K: results per retriever in the pool L: results per retriever fed to RRF

Both depths change together: L = K



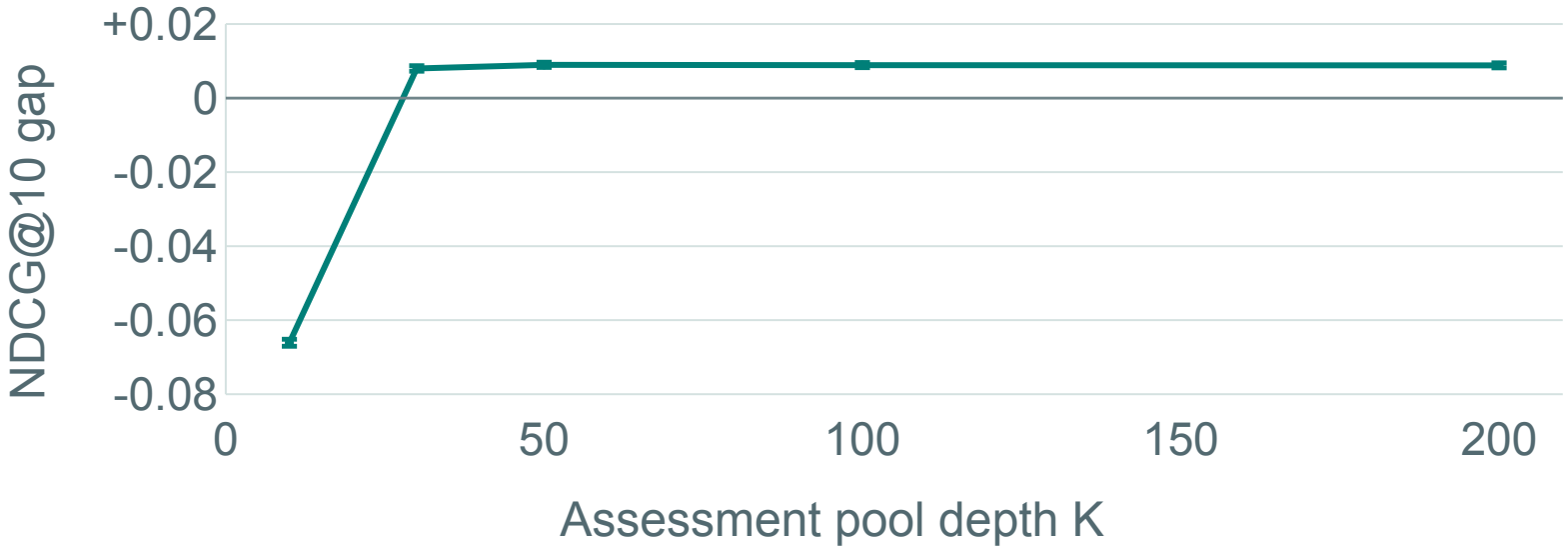
At K = 200: a small gap of +0.0088 in pooled NDCG@10

Pseudo-labels from one judge. 95% paired CI [0.0082, 0.0095].
Query-bootstrap intervals exclude judge uncertainty.

Pool coverage can reverse a comparison

K: results per retriever in the pool L: results per retriever fed to RRF

Only K changes. L = 200 and both rankings stay fixed.



14.4%

of RRF top-10 pairs outside the pool at K = 10

145,643 query–passage pairs

K = 10: −0.0661

K = 30: +0.0080

RRF minus E5 under pseudo-labels. Unjudged passages receive zero gain.

Scope and robustness

Results survive query exclusions

Evaluation	Remaining queries	Observed result
Manual labels 23 affected queries excluded	1,156	System order unchanged
Full-set pseudo-labels Manual queries excluded	99,914	Signs of RRF–E5 differences unchanged

Manual pools: reference-assisted, with unmeasured assessor agreement.

Large pools: one judge, without independent human validation.

Generalization beyond selected English MS MARCO pools remains untested.

Judgment coverage affects retrieval evaluation

Human judgments

E5 NDCG@10: 0.488 to 0.879

Same ranking, more complete relevance judgments

Fixed rankings, one automatic judge

Pool coverage can reverse the RRF vs E5 comparison

Next: test stronger judges and known-answer inputs
against independent human judgments.

Evidence from selected English MS MARCO candidate pools.

NDCG@10 and paired annotation effects

NDCG@10 = DCG@10 / ideal DCG@10

Ranking	Human − original	95% paired CI
BM25	+0.396	0.372 to 0.419
E5	+0.391	0.365 to 0.417
BGE	+0.395	0.369 to 0.419
RRF (3)	+0.395	0.371 to 0.419

5,000 paired query bootstraps. All four annotation effects: $p < 0.001$.

Judge test protocol and observed errors

Train queries

825

Validation queries

177

Test queries

177

Test target	Predicted negative	Predicted positive
Coded negative	1,300	506
Coded positive	314	2,882

Precision 0.851 **Recall 0.902** **F1 0.875**

5,002 pairs; six unresolved targets mapped to zero and one empty-text record.

Full-set depth results

K	Candidates / query	Predicted relevant / query	E5 NDCG@10	RRF L=K NDCG@10	RRF L=200 NDCG@10
10	21.9	14.2	0.871	0.852	0.805
30	65.8	33.7	0.851	0.862	0.859
50	110.4	49.4	0.847	0.856	0.856
100	222.9	81.0	0.844	0.851	0.853
200	451.5	128.6	0.842	0.850	0.850

Single-judge pseudo-labels. All 101,093 queries remain in every average.

Sensitivity to query exclusions

Human-label NDCG@10

Ranking	All 1,179 queries	1,156 after exclusions
BM25	0.644	0.644
E5	0.879	0.879
BGE	0.828	0.828
RRF (3)	0.854	0.855

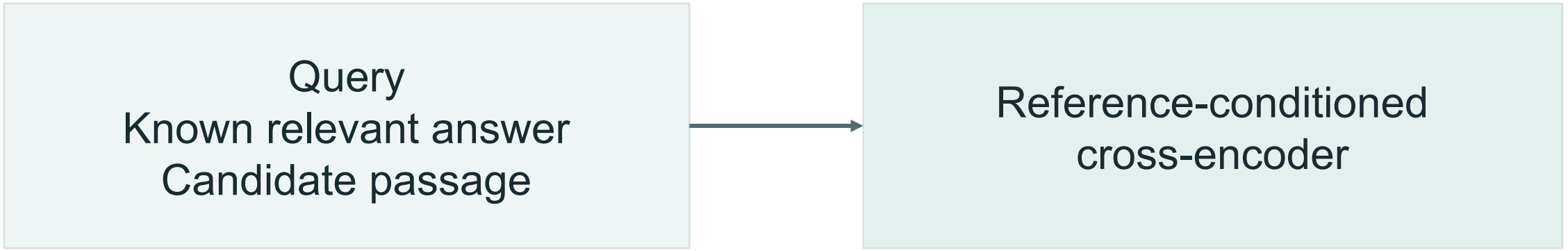
Full set without manual queries: N = 99,914

At K = 200: RRF − E5 = +0.00877

Exclusion checks do not replace independent human assessment.

Next validation: reference-conditioned judging

Proposed input



Compare stronger judges against independent human assessments.

Planned evaluation. No results for this model are reported in the present study.